



भारतीय भूवैज्ञानिक सर्वेक्षण GEOLOGICAL SURVEY OF INDIA

जम्मू एवं कश्मीर के रियासी जिला के अंतर्गत श्री माता वैष्णो देवी हिमकोटी ट्रैक पर भूस्खलन एवं
ढलान स्थिरता के आकलन पर प्रतिवेदन ।

(मदकोड: एम 4-एलएस / एनसी / एनआर / एसयू-जेके / 2017/12224)

(फील्डसीजन: 2017-18)

REPORT ON THE LANDSLIDE AND SLOPE STABILITY ASSESSMENT OF SHRI MATA VAISHNO DEVI HIMKOTI TRACK, REASI DISTRICT, JAMMU AND KASHMIR.

(Item Code: M4-LS/NC/NR/SU-JK/2017/12224)

(Field Season: 2017-18)

द्वारा

देवेन्द्र सिंह वरिष्ठ भूवैज्ञानिक
सुरेश कुमार राणा वरिष्ठ भूवैज्ञानिक
संजय जंगराल वरिष्ठ भूवैज्ञानिक

By

Devender Singh, Senior Geologist
Suresh Kumar Rana, Senior Geologist
Sanjay Jungral, Senior Geologist

अभियांत्रिकी भूविज्ञान एवं भूस्खलन प्रभाग
Engineering Geology and Landslide Division
मिशन IV, उत्तरी क्षेत्र / Mission- IV, Northern Region
राज्य इकाई जम्मू एवं कश्मीर
State Unit: Jammu & Kashmir
जम्मू/Jammu

2018

प्रतिवेदन की साफ्ट प्रति एवं आंकिक आंकड़ें महानिदेशक, भारतीय भूवैज्ञानिक सर्वेक्षण, कोलकाता-700016 की अनुमति से प्राप्त
किये जा सकते हैं।

Soft copy of the report and digital data may be obtained with the permission of Director General,
Geological Survey of India, Kolkata -700016



भारतीय भूवैज्ञानिक सर्वेक्षण GEOLOGICAL SURVEY OF INDIA

जम्मू एवं कश्मीर के रियासी जिला के अंतर्गत श्री माता वैष्णो देवी हिमकोटी ट्रैक पर भूस्खलन एवं
ढलान स्थिरता के आकलन पर प्रतिवेदन ।

(मदकोड: एम 4-एलएस / एनसी / एनआर / एसयू-जेके / 2017/12224)

(फील्डसीजन: 2017-18)

REPORT ON THE LANDSLIDE AND SLOPE STABILITY ASSESSMENT OF SHRI MATA VAISHNO DEVI HIMKOTI TRACK, REASI DISTRICT, JAMMU AND KASHMIR.

(Item Code: M4-LS/NC/NR/SU-JK/2017/12224)

(Field Season: 2017-18)

द्वारा

देवेन्द्र सिंह वरिष्ठ भूवैज्ञानिक
सुरेश कुमार राणा वरिष्ठ भूवैज्ञानिक
संजय जंगराल वरिष्ठ भूवैज्ञानिक

By

Devender Singh, Senior Geologist
Suresh Kumar Rana, Senior Geologist
Sanjay Jungral, Senior Geologist

अभियांत्रिकी भूविज्ञान एवं भूस्खलन प्रभाग
Engineering Geology and Landslide Division
मिशन IV, उत्तरी क्षेत्र / Mission- IV, Northern Region
राज्य इकाई जम्मू एवं कश्मीर
State Unit: Jammu & Kashmir
जम्मू/Jammu

2018

प्रतिवेदन की साफ्ट प्रति एवं आंकिक आंकड़ें महानिदेशक, भारतीय भूवैज्ञानिक सर्वेक्षण, कोलकाता-700016 की अनुमति से प्राप्त
किये जा सकते हैं।

Soft copy of the report and digital data may be obtained with the permission of Director General,
Geological Survey of India, Kolkata -700016

**REPORT ON THE LANDSLIDE AND SLOPE STABILITY ASSESSMENT OF SHRI
MATA VAISHNO DEVI HIMKOTI TRACK, REASI DISTRICT, JAMMU AND
KASHMIR.**

**(Item Code: M4-LS/NC/NR/SU-JK/2017/12224)
(Field Season: 2017-18)**

By

**Devender Singh, Senior Geologist
Suresh Kumar Rana, Senior Geologist
Sanjay Jungral, Senior Geologist
SU: Jammu & Kashmir
Geological Survey of India, Northern Region, Jammu**

CONTENTS

	Page No.
सारांश	i
Abstract	ii
I. INTRODUCTION	1
1.1 General	1
1.2 Objective and scope of the present study	1
1.3 Location	2
1.4 Physiography	2
1.5 Climate, rainfall and drainage	3
1.6 Flora and fauna	4
1.7 Previous Work	4
1.8 Present Work	4
Acknowledgements	4
II. GEOLOGY	
REGIONAL GEOLOGY	7
GEOLOGY OF THE STUDY AREA	9
III. GEOTECHNICAL APPRAISAL	12
IV. CONCLUSION	13
V. RECOMMENDATIONS	14
Locality Index	15
References	16

List of Tables

Table-1: Geology of the study area.	6
Table-2: Joint characteristics measured at site.	6

List of Figures

Figure 1: Bird's eye view and location map of slide Area showing vulnerable site	
Figure 2: DEM image of the study area.	
Fig.3: Compile Geological map of part of Toposheet No.43L/13 & 43K/16.	
Figure 4: Cross cutting tight joints in dry Condition.	
Figure 5: The view of the damage caused by slide to shed and track.	
Figure 6: Stereo plot of the slope and discontinuity planes.	
Figure 7: 3D view of discontinuity planes.	
Figure 8: Sketch geological map of affected site.	
Figure 9: Schematic cross section along X-Y line.	

जम्मू एवं कश्मीर के रियासी जिला के अंतर्गत श्री माता वैष्णो देवी हिमकोटी ट्रैक पर भूस्खलन एवं
ढलान स्थिरता के आकलन पर प्रतिवेदन ।

(मदकोड: एम 4-एल एस / एन सी / एन आर / एस यू-जेके / 2017/12224)

(फील्ड सीजन: 2017-18)

द्वारा

देवेंद्र सिंह, वरिष्ठ भूवैज्ञानिक

सुरेश कुमार राणा, वरिष्ठ भूवैज्ञानिक

संजय जंगराल, वरिष्ठ भूवैज्ञानिक

राज्य इकाई: जम्मू एवं कश्मीर

भूवैज्ञानिक सर्वेक्षण भारत, उत्तरी क्षेत्र, जम्मू

सारांश

श्री वैष्णवों देवी स्थल के हिमकोटी ट्रैक रोड पर पहाड़ी ढलान में अतीत में अस्थिरता देखी गयी है, जो तीर्थ यात्रियों के लिए दुर्घटना का कारण बनती है। अतिरिक्त मुख्य कार्यकारी अधिकारी, श्री माता वैष्णो देवी श्राइन बोर्ड, कटरा, के अनुरोध पर भारतीय भूवैज्ञानिक सर्वेक्षण द्वारा आर डी 0/850 किलोमीटर और कुछ अन्य चयनित हिस्सों में ढलान का भू-तकनीकी मूल्यांकन किए गए हैं।

परियोजना प्राधिकरणों की आवश्यकता के अनुसार प्रस्तावित साइट पर ढलान विफलता के कारण का पता लगाने के लिए भू तकनीकी जांच की गई।-जांच के दौरान, यह पाया गया कि प्रभावित क्षेत्र में संरचनात्मक अंतरालों की उपस्थिति, जो कि पहाड़ी की ढलान के समरूप है, ने इस क्षेत्र को कमजोर बना दिया है।

क्षेत्र में भारी बारिश के कारण संधि सतहों के माध्यम से पानी का व्यापक रिसाव होता है और इससे उत्पन्न चिकनाहट से उन विशेष स्थानों पर भूस्खलन होता है जिससे जीवन और संपत्ति को नुकसान पहुँचता है।

**REPORT ON THE LANDSLIDE AND SLOPE STABILITY ASSESSMENT OF SHRI MATA
VAISHNO DEVI HIMKOTI TRACK, REASI DISTRICT, JAMMU AND KASHMIR.**

**(Item Code: M4-LS/NC/NR/SU-JK/2017/12224)
(Field Season: 2017-18)**

By

**Devender Singh, Senior Geologist
Suresh Kumar Rana, Senior Geologist
Sanjay Jungral, Senior Geologist
SU: Jammu & Kashmir
Geological Survey of India, Northern Region, Jammu**

ABSTRACT

The mountainous slope in the Himkoti Track road has reported to have experienced instability in recent past which lead to casualty to pilgrims bound to shrine. On the request of Additional Chief Executive Officer, Shri Mata Vaishno Devi Shrine Board, Katra, GSI has carried out geotechnical assessment of the slope at RD 0/850 km and at some other selected stretches. The geotechnical investigation was carried out to ascertain the cause of slope failure at proposed site as per the requirement of project authorities. During the investigation, it was found that the presence of structural discontinuities, in the affected area, dipping toward the valley side following the hill slope (dip slope) has created this stretch vulnerable. Heavy rainfall in the area resulting into widespread seepage of water through joint planes and lubricating them leading to landslide at particular spot causing damage to life and property.

REPORT ON THE LANDSLIDE AND SLOPE STABILITY ASSESSMENT OF SHRI MATA VAISHNO DEVI HIMKOTI TRACK, REASI DISTRICT, JAMMU AND KASHMIR.

(Item Code: M4-LS/NC/NR/SU-JK/2017/12224)

(Field Season: 2017-18)

By

Devender Singh, Senior Geologist

Suresh Kumar Rana, Senior Geologist

Sanjay Jungral, Senior Geologist

SU: Jammu & Kashmir

Geological Survey of India, Northern Region, Jammu

INTRODUCTION

1.1 General

The holy shrine of Shri Mata Vaishno Devi is one of the famous ancient pilgrimage centers of the Hindus located at the Trikuta Hills within the State of Jammu and Kashmir.

The mountainous slope in the Himkoti Track road has reported to have experienced instability in recent past which lead to casualties of shrine bound pilgrims. The Additional Chief Executive Officer, Shri Mata Vaishno Devi Shrine Board, Katra vide his letter no. CO/Dev/113/1800-1801, dated 30.06.2017 has requested GSI for an assessment of the slope at RD 0/850, km. In view of the above and as per the direction of the Supervisory officer EG & NLSM, GSI, SU:J&K, Jammu, the preliminary landslide and slope stability assessment of the Bhawan area has been taken up under the approved Annual Field Season program of Geological Survey of India, FS: 2017-18, Item code:M4-LS/NC/NR/SU-JK/2017/12224. A comprehensive geotechnical investigation was carried out to ascertain the cause of slope failure at site as per the requirement of project authorities.

The GSI team comprises of S/Shri Devender Singh and Suresh Kumar Rana Sr. Geologists of EG division carried out the onsite geotechnical assessment of the slope.

1.2 OBJECTIVE AND SCOPE OF THE PRESENT STUDY:

The study was taken-up with the objective to ascertain the stability of slope above RD 0/850 km of Himkoti Track (Fig. 1&5) which is protruding and appeared prima facie a potential rock fall zone and to suggest appropriate remedial measures for the stability and safety of pilgrims.

1.3 LOCATION:

Katra or Katra Vaishno Devi, as it is popularly known, is a small town in Reasi district of Jammu and Kashmir situated in the foothills of the Trikuta Mountains, where the holy shrine of Vaishno Devi is located. Vaishno Devi cave near Katra is an important shrine for the Hindus and the most famous tourist attraction of this district. The Latitude and Longitude of the investigated site is $33^{\circ}01'47.5''\text{N}$ and $74^{\circ}56'8''\text{E}$ (Fig. 2).

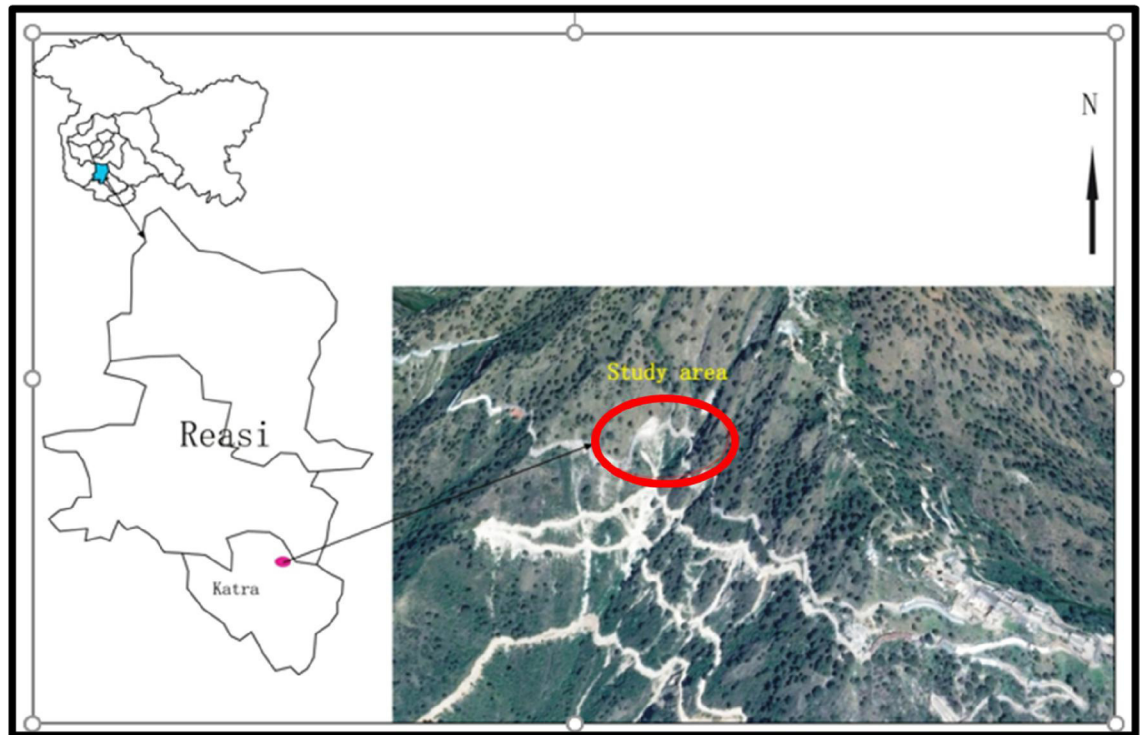


Fig.1. Bird's eye view and location map of slide Area showing vulnerable site.

1.4 PHYSIOGRAPHY

The area is characterized by very rugged topography having high hills and deep valleys with precipitous slopes. The Sirban Limestone stands out prominently with high pointed, tapering peaks and sub-vertical slopes because of dolomitic and silicious lithology, resistant in nature in comparison to the Tertiary rocks in vicinity. The Trikuta peak (2384m) is a prominent feature in the area.

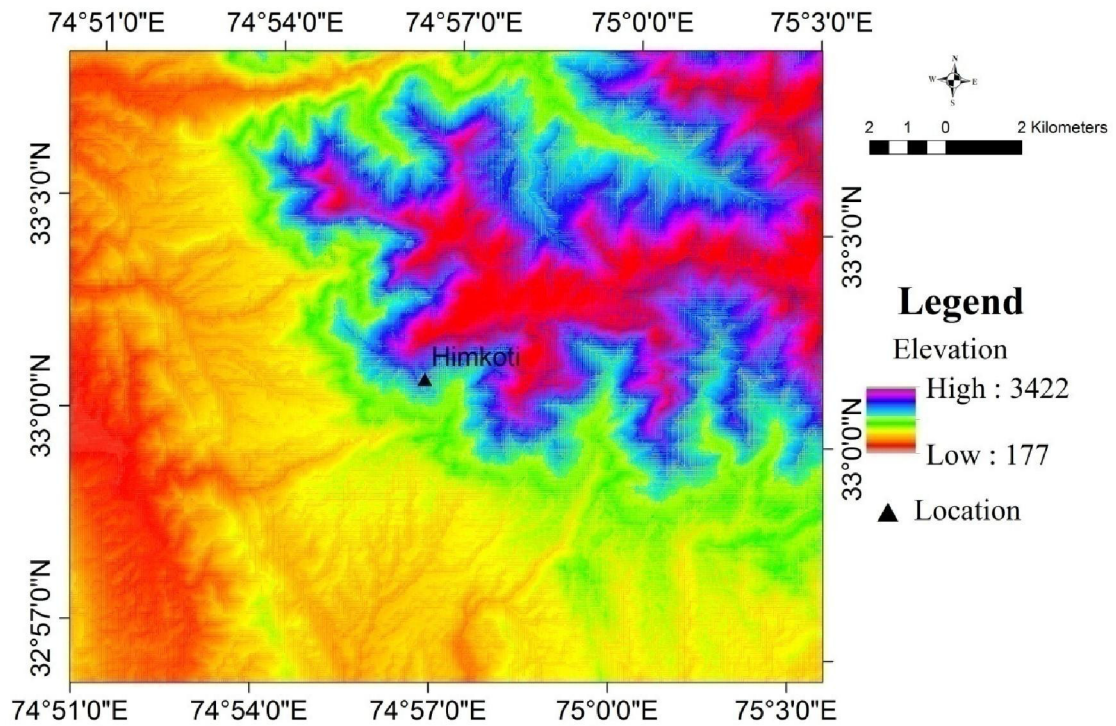


Fig.2: DEM image of the study area.

1.5 CLIMATE, RAINFALL AND DRAINAGE

The area falls in sub-tropical to temperate climate zone and witnesses moderate to extreme climate characterized by hot summers and cold winters. The higher reaches experience extreme cold and occasional snowfall during winter months. The rainy season is from July to September. The area experiences average annual rainfall of 1200 mm. The regional drainage of the area is Chenab River, Tawi River and their tributaries. The present study area is mainly drained by by *Pei Khad* and *Balganga nala* tributaries of the Chenab River. Except rainy seasons, when these are fully charged, these tributaries generally remain dry during the year. Sub-surface drainage is also a peculiar character of the area being dolomitic in nature.

1.6 FLORA AND FAUNA

The area is covered by Chir (*Pinus roxburghii*) and broad leaf trees along the mountain slopes of higher altitudes whereas escarpment faces are generally devoid of much vegetation (due to little soil cover) except thin and patchy growth of pine trees. The lower reaches support a mix type of vegetation, i.e. Chir (*Pinus roxburghii*), Amla (*Dodonea viscosemblica*), Garna (*Carrisa spinarum*), Anardana (*Sourpomegranate*), Mango (*Mangifera indica*), Kiker (*Acacia arabica*), Sisham (*Dalbergia sissoo*), Ber (*Zizyphus jujoba*)

,Cactus and numerous varieties of bushes and shrubs and medicinal plants. The tall grasses also grow in the area. The fauna include bear, leopard, wild fowl, monkey, variety of snakes and scorpion. The domestic animals are buffalo, cow, horse, sheep and goat.

1.7 PREVIOUS WORK

Chauhan and Mallick (1983) prepared a hazard zonation map of the area. Ajay Kumar carried out geotechnical investigation of landslide near Bhawan and suggested various remedial measures for the wedge failure and sliding/detachment, which had taken place in the densely inhabited areas and foot tracks for the pilgrims to the Holy Shrine of Shri Mata Vaishno Devi and Bhairon Baba Temple.

Besides geotechnical investigations Vaishno Devi Limestone has been explored for various reasons by many workers. It was Medlicott (1876) who first studied these inliers and named them as “Great Limestone”. Subsequently these inliers have been variously designated as “Infra-Trias Limestone” (Middlemiss, and Wadia, 1937), “Sirban Limestone” Middlemiss, 1896, Wadia 1937), “Jammu Limestone” (informally by Wadia, 1937) and “Raisi Limestone” (locally and informally by various workers).

1.8 PRESENT WORK

This report embodies work carried out as preliminary landslide investigation started on 1st July, 2017 for six days, to ascertain the causes of the landslide at RD 0/850 on Himkoti track slope and appropriate remedial measures suggested to concerned authorities at site.

ACKNOWLEDGEMENTS

The authors express their sincere thanks to Shri Som Nath Chandel, ADG & HOD and Smt. Varsha Ashok Aglawala, Dy. D. G., RMH-IV, NR, Lucknow. Shri D. V. Ganvir, Deputy Director General, SU: J&K, Jammu for their valuable guidance. Special thanks to Shri Vijay Shivgotra, Director, Engineering Geology& Landslide Division, SU: J&K, Jammu for his valuable technical guidance during fieldwork. The authors are thankful to the Dy.CEO and the authorities of Shri Mata Vaishno Devi Shrine Board (SMVDSB) for providing necessary permission to carry out fieldwork as well as for extending necessary support during the course of fieldwork.

II. GEOLOGY

REGIONAL GEOLOGY

The studied area is part of the Trikuta hills exposing Meso-Proterozoic Sirban Group. Dolomite/ cherty dolomite and flaggy limestone units are exposed in and around the studied area. These rocks are at places sheared, fractured & jointed. The dolomite is white to off-white, hard, compact and fine grained which at places shows well developed columnar stromatolites. Bedding is marked by laminations and colour bandings. Tanhal Thrust and Reasi Thrust are the two major structural discontinuities present as regional tectonic features in the area and are exposed about 8 to 10 km in the south from Vaishno Devi Shrine (Fig. 3).

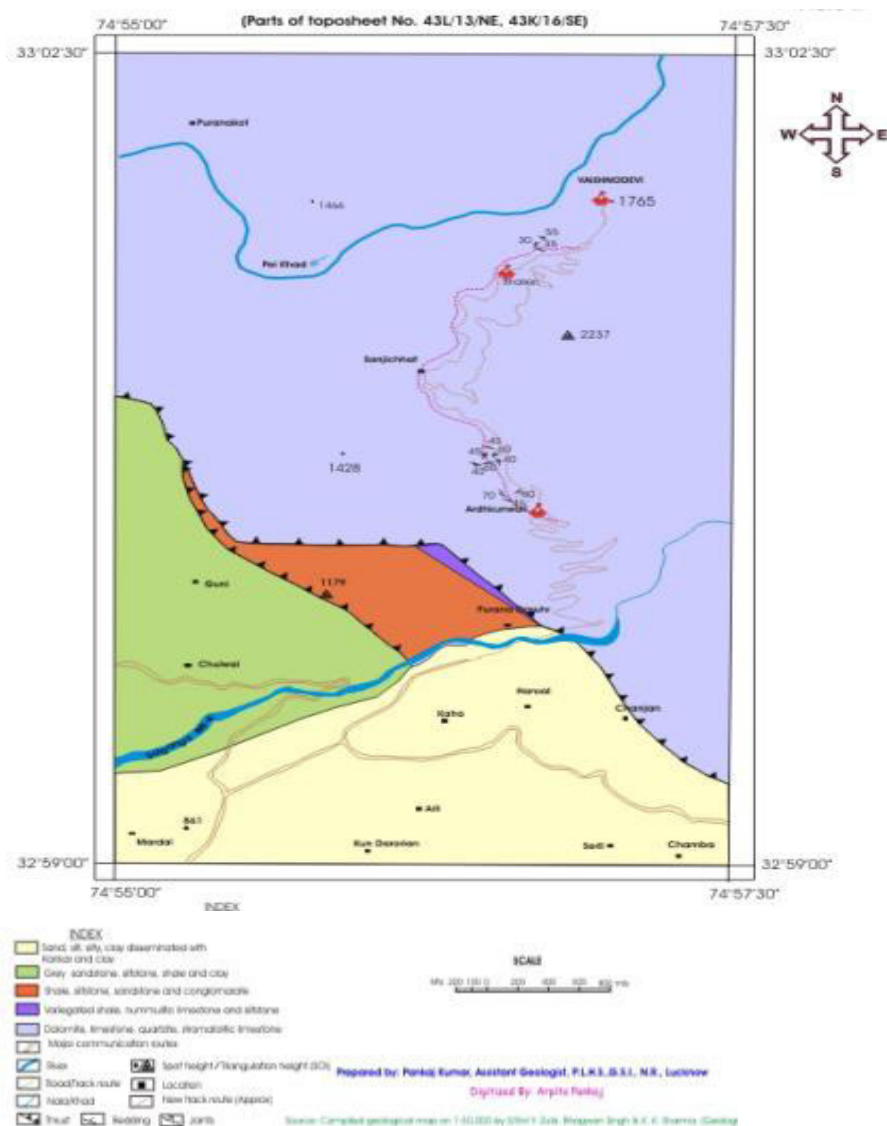


Fig.3: Compile Geological map of part of Toposheet No.43L/13 & 43K/16.

The lithological succession and assemblages of Sirban Group is as follows:

Table-1: Geology of the study area (after Gupta, 1997):

Group	Formation	Lithology	Age
Sirban	Khairikot Formation	Variegated shale, black shale/slate with quartzite lenses. Quartzite beds with breccia of sub-angular to angular pebble of chert, and dolomite/limestone	Meso-Proterozoic
	Trikuta Formation	Dolomite with stromatolite, flaggy limestone with red- chert, cherty to non cherty dolomite with thin bands of stromatolites. Flaggy limestone with slate partings and Dolomite with stromatolites.	

The hill slope at the affected site is steep, however, locally found to be sub-vertical. It presents a mean slope inclination of the order of 67°. At places, negative slopes have also been recorded (Fig.1). The slope forming material comprises dolomitic limestone and cherty dolomite. The rock mass at RD 0/850 km of Himkoti Track is found to be traversed by 3 plus random sets of joints which are variably-spaced, dry and tight in nature and (Fig. 4; Table 2).

Table-2: Joint characteristics measured at site:

Sl. No	Joint Number	Strike & Dip	Continuity (m)	Spacing (cm)	Opening	Remarks
1	J1 (Bedding joint)	N60°W-S60°E/40° NE	200 - 250	70-100	2 - 3 mm	Smooth
2	J2	N40°E-S40°W/85°SE	20 – 25 m	50-75	Very tight	Smooth
3	J3	N55° E-S55°W/60°NW	30 – 35 m	5-10	Tight	Slightly rough
4	J4(random joint)	N25° E-S25°W/60°NW	< 1 m	100-150	Tight	Slightly rough



Fig.4: Cross cutting tight joints in dry Condition.

GEOLOGY OF THE SYUDY AREA

Rocks belonging to Sirban Group represented by dolomite/cherty dolomite are exposed in and around the area of investigation. Outcrops of dolomite /cherty dolomite occupy major part of the area. The rock mass is traversed by three major and one random joint set. These joint sets are both open and tight. Solution effects along joint planes and shears zones and along bedding planes are common in Sirban Group. Staining along the joints has also been recorded in the area.



Fig.5: The view of the damage caused by slide to shed and track.

A major tectonic plane Reasi Thrust passes through a few hundred meters west of the study area along which the older Sirban Group of rocks tectonically overlies the younger Siwalik Group of rocks comprising sandstone, siltstone and mudstone. The general trend of the rocks is N60°W-S60°E with dip varying from 40° to 60° north-easterly. The joints recorded in the rock mass are given in Table-2. Secondary filling, mainly calcareous along the joint planes have also been recorded in the area. Minor shear zones parallel and transverse to the bedding plane have also been recorded in the area.

The height difference between these unstable blocks and the Track is about 100 m. The overall slope angle 67° southerly. This is high rainfall area and seismically very active zone with zone factor of 0.36. The chronic instability of the hill is around 55 %.(VE). The damage corridor width is 50m (Fig.5). Loose rock/potential unstable materials, which have moved out of place on the slope, are noticed. Hence it is recommended to manually remove the loose and detached rock mass upslope.

III. GEOTECHNICAL APPRIASAL

Based on the geometric orientations of four joint planes (Table. 2), a kinematic admissibility analysis is carried through stereographic projection of the measured joint planes and the prevalent topography (Fig.6, 7). The stereographic plot indicates that the major wedges or topple mode of failures is plausible at the prevalent slope face because of orientation of J1, J2 and J3. The J2 is to be perpendicular to bedding and the most predominant and pervasive discontinuity, which however, is dipping unfavourably or towards track. The J2 failure plane "daylights" on the slope face. Although J4 is measured as a random set but locally, it has the potential to form planner/topple mode of failures towards the prevalent slope directions. Allowing further failure at this location will pose great danger to the pilgrims and Track.

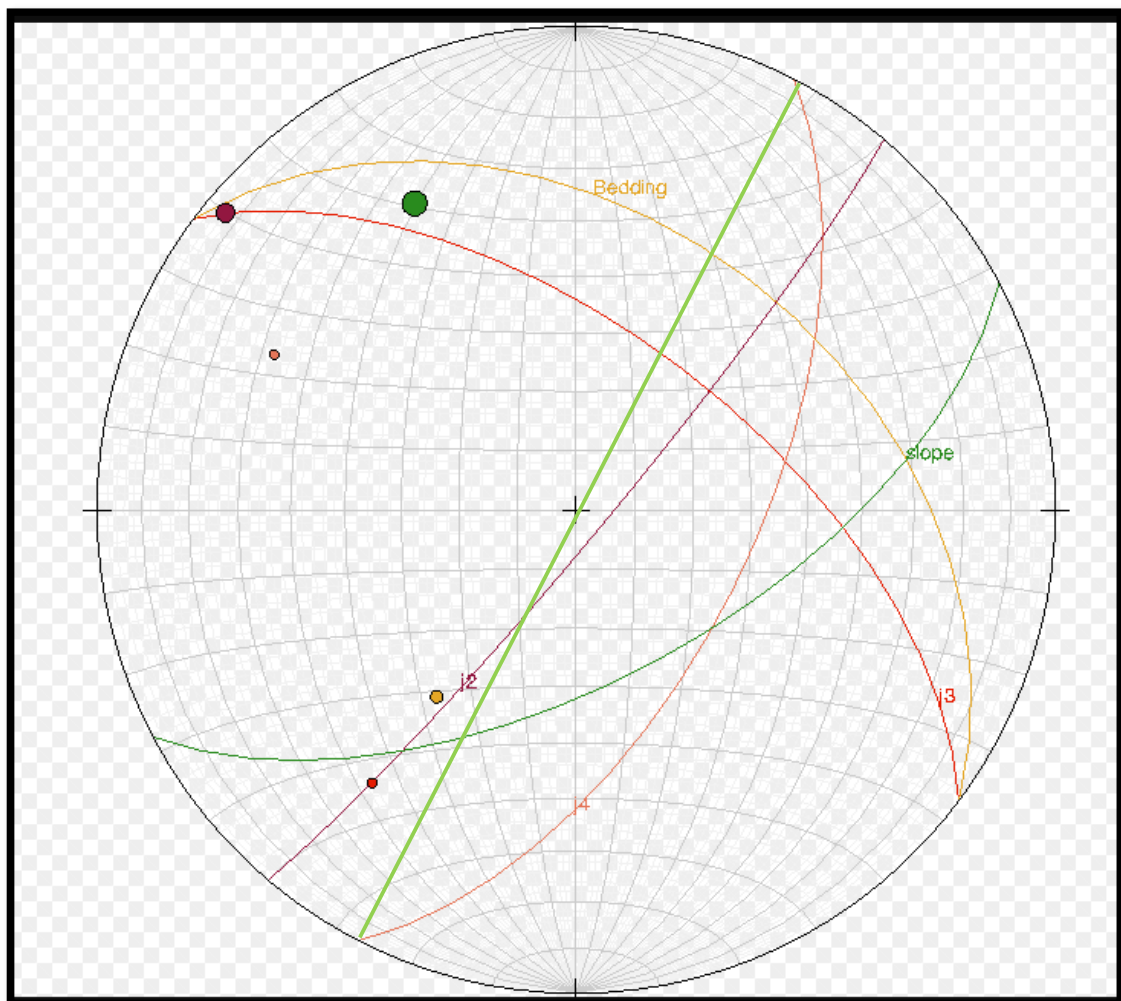


Fig.6. Stereo plot of the slope and discontinuity planes

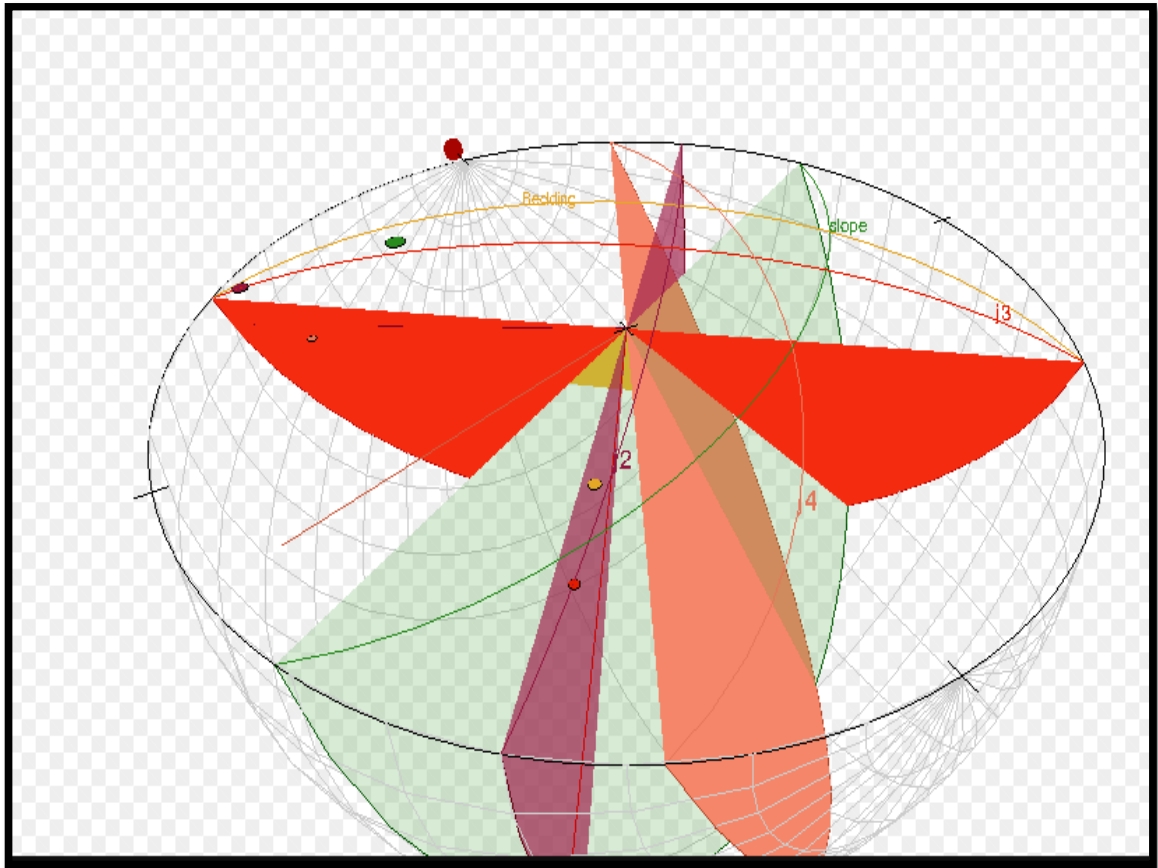


Fig.7: 3D view of discontinuity planes.

Based on the above assessment and to prevent any possible movement of failed rock blocks due to wedge/ topple failure rock barrier fencing may be installed across the projected run out zone (Fig. 8,9). The rock barrier fencing may be of with high tensile cables with tensile strength of at least $1,770 \text{ N/mm}^2$. The height of these rock barrier fences and strengths of the steel cables need to be designed so that the same can withstand the weight and momentum of large size failing rock blocks (maximum boulder size observed at site is 0.5 to 2m approximately), including taking up some drainage improvement work so that water is not seeped into the joints and fractures at the potential rock fall zone. The cactus trees can be removed at potential slopes, which are responsible for mechanical breaking of rock mass along upslope.

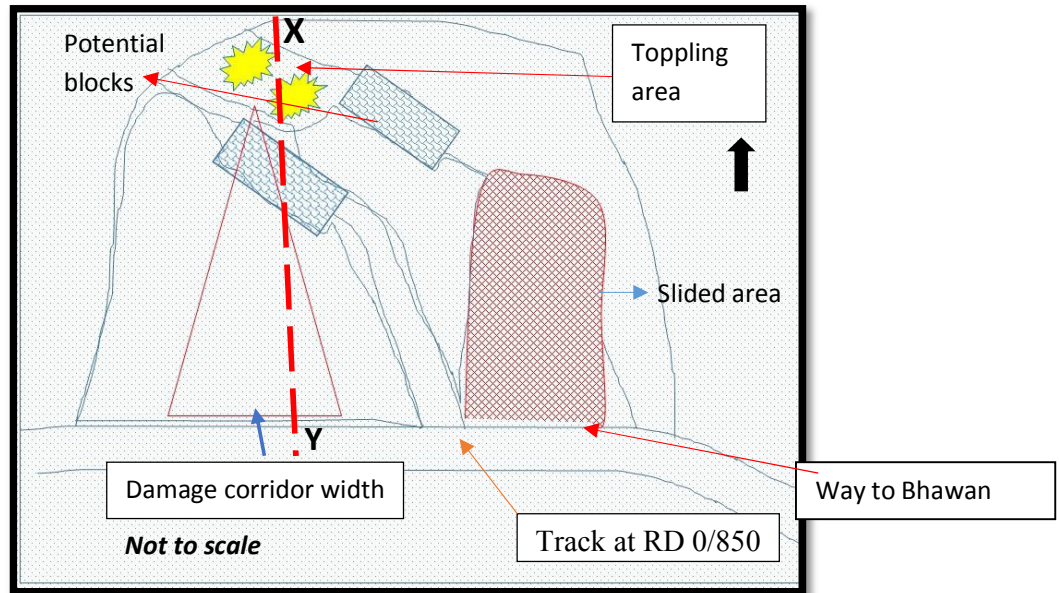


Fig.8. Sketch geological map of affected site.

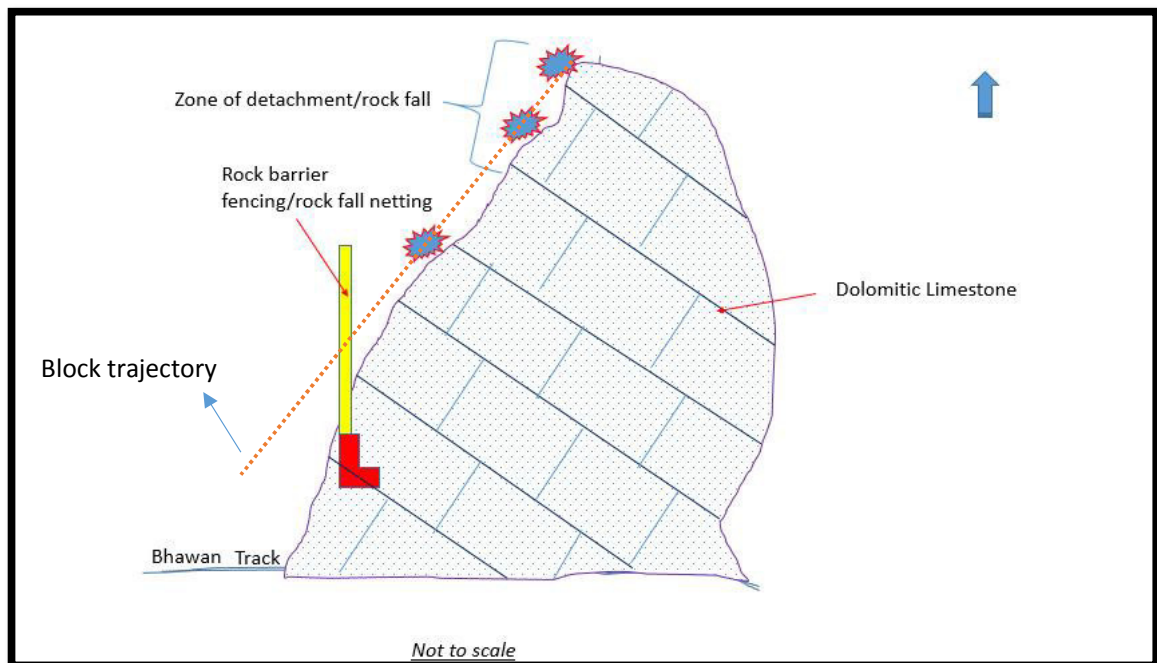


Fig.9: Schematic cross section along X-Y line.

IV. CONCLUSION

1. The Damage Corridor Width of landslide behind the existing slope varies in width from 30 m to 50 m.
2. The rocks of the local slope at this location consist of highly jointed dolomitic limestone.
3. The slopes at this location are found prima-facie unstable during various analyses. Topple and wedge failures are main modes of failure.

V. RECOMENDATIONS

Based on the preliminary geotechnical assessment of RD 0/850 on Himkoti track slope following remedial measure have been suggested:

1. Nets/ rock fall netting and meshes made from high-tensile steel wire with tensile strength of at least 1,770 N/mm² /Maximum Energy Level (MEL) should be placed with hinged support at slope considering the size of boulders which are slid down in the area.
2. The barrier has to catch at least two impacts of a boulder with 1/3 of the MEL energy without damage. The residual height after the first impact must be greater than 70%.
3. The high tensile strength of the cables/rock fall netting to be designed considering the site conditions, maximum size of boulders are 5meter. (Fig.5,9) and height according to trajectory.
4. Loose rock/potential unstable materials, which have been displaced/dislodged from insitu rock needs to be removed at regular intervals.
5. Detailed study is required to identify the areas prone to sliding all along the track, particularly, upslope to suggest suitable slope protection measures on 1:1000 scale.

LOCALITY INDEX

Locations	Latitude	Longitude
Ardhkwari	33°00'40"	74°56'40"
Bhawan	33°01'41"	74°56' 42"
Katra	32° 59' 30"	74° 56' 00"
Reasi	33° 04'50"	74° 49'40"

REFERENCES

(After Gupta 1997) Geology of the study area.

Burnett, 1975: Determining intact rock strength.

Chauhan, A., Sharma S. K., 2015: Report on Preliminary geotechnical assessment on landslide investigation along Shri Mata Vaishno Devi Shrine Board Track, Reasi district, Jammu and Kashmir.

Dr. Read SAL, 2005: Field description of Soil and Rocks.

Giovanni Vaciago: Engineering measures for Slope stabilisation.

Google earth, 03/07/2017: Image of Area.

Kumar, S.V, Kumar Ajay, 2008: A report on the geotechnical investigation of various components in Ardhkwari temple complex, district Reasi, Jammu & Kashmir.

Maccaferri barrier design for rock fall.

Medlicot (1876) : Geotechnical investigation of Vaishno Devi Limestone.

Middlemiss and Wadia, 1973: Sirban Limestone.

Middlemiss, 1896 and Wadia, 1973: Jammu Limestone.

Wadia, 1937: Raisi Limestone.

DATA SHEET FOR PRELIMINARY INVESTIGATION OF LANDSLIDE

No.	Field	Description.
1	Slide no. :	JK/Katra/43K/13& 16/RD 0/850
2	State :	Jammu and Kashmir.
3	District :	Reasi
4	Toposheet :	Part of 43L/13& 43K/16.
5	Name of the slide :	RD 0/850
6	NH/SH/Locality :	Himkoti
7	Latitude :	33°01'47.5"N
8	Longitude :	74°56'8" E
9	Length :	
10	Width :	

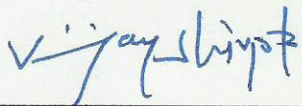
11	Height :	
12	Area :	
13	Depth :	
14	Volume :	
15	Run out Distance :	
16	Type of material :	
17	Type of movement :	Slide.
18	Rate of movement :	
19	Activity :	
20	Distribution :	
21	Style :	Planner failure.
22	Failure mechanism :	Repeated sliding along planes.

23	History :	Initiated on 1st July, 2017
24	Geomorphology :	The topography of area is rugged and mountainous with varying altitudes ranging from 868m (Balganga), 1048m (Charan Paduka), 1482m (Ardhkunwari), 2041 m (Sanjichat), 1900m (Bhawan) to 2200m (Bhairon). The Sirban limestone forms the high ridge trending WNW-ESE with steep slopes while the younger rocks (Tertiary) occupy the low topography with gentle slopes. The drainage is mainly governed by the Balganga and Pei Khad rivers and shows dendritic pattern. Number of fresh water springs occurs in the area.
25	Geology :	Dolomite, slate, quartzite and stromatolitic dolomite of Trikuta formation of Sirban group having Meso-Proterozoic.
26	Structure :	The regional structure of the area displays the Reasi-Katra inlier as a southerly plunging anticline with Sirban Group in the core and younger Tertiaries fringing on it. The Sirban Group is thrust over the younger rocks along this fault.
27	Landuse/ Landcover :	Agricultural and settlement/ Shelter (Belongs to SMVDSB).
28	Hydrological condition :	Dripping.
29	Triggering factor :	Heavy rainfall.
30	Death of persons :	
31	People affected :	Yatris/Pedestrians' partially affected due to safety.
32	Livestock loss	Nil.

	:	
33	Communication :	Road damaged.
34	Infrastructure :	Buildings/sheds. (as per locals)
35	Agriculture/forest/barren :	Forest
36	Geo-scientific Causes :	
37	Remedial measures :	1. Nets/ rock fall netting and meshes made from high-tensile steel wire with tensile strength of at least 1,770 N/mm² /Maximum Energy Level (MEL) should be placed with hinged support at slope considering the size of boulders which are slid down in the area. 2. The barrier has to catch at least two impacts of a boulder with 1/3 of the MEL energy without damage. The residual height after the first impact must be greater than 70%. 3. Loose rock/potential unstable materials, which have been displaced/dislodged from insitu rock needs to be removed at regular intervals.
38	Remarks :	-
39	Photos. Sketch of plan	-

	§ion of the slide :	
40	Summary/Abstract :	-
41	Pdf :	-

FIVE POINT PROFORMA

Title of the report	Report on the landslide and slope stability assessment of Shri Mata Vaishno Devi Himkoti Track, Reasi District, Jammu and Kashmir.
Name(s) of the author(s)	Devender Singh, Senior Geologist Suresh Kumar Rana, Senior Geologist Sanjay Jungral, Senior Geologist
Background of the investigation at whose instance or with what particular objective in view the investigation was undertaken	Since GSI is a nodal agency for landslides and as per guidelines by NMH-IV, by formal approval by HOD, NR, the Landslide investigation is initiated.
Whether the item of investigation was included in the approved programme for the year concerned. If so, particulars, such as item No., page reference etc. may be given	(Item Code: M4-LS/NC/NR/SU-JK/2017/12224) (Field Season: 2017-18)
Date of commencement of the investigation with month and year, and exact number of days spent by the officer(s) in the field in connection with the investigation	01-07-2017 Devender Singh, Senior Geologist (06 days) Suresh Kumar Rana, Senior Geologist (06 days) Sanjay Jungral, Senior Geologist (06 days)
Brief comments by Director / Supervisory officer on the conclusion and recommendation by the author(s) with suggestion for implementing the same.	The conclusions and recommendations were made based on synthesis of field data.
Please state specifically how the results of the investigation were achieved compared with the objective for which it was undertaken	The results were achieved based on field data collection and interpretation.
Signature of the Project Director: 	Name: Vijay Shivgotra Designation: Director Project/ Division: EG & Landslide State Unit/ Region: Jammu & Kashmir, NR
Place Date	Jammu 26-06-2018



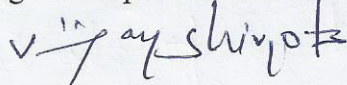
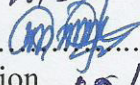
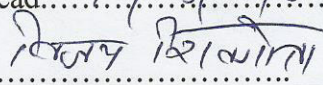
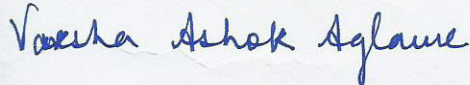
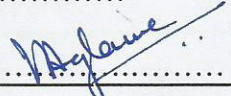
Geological Survey of India

Data sheet on report movement in connection with its scrutiny

Division: Engineering Geology and Landslide, FSP No: M4-LS/NC/NR/SU-

JK/2017/12224, Mission: IV

Title: A note on the Landslide and Slope stability assessment of Shri mata Vaishno Devi Himkoti track, Reasi district, Jammu and Kashmir.

Name of author(s) Devender Singh, Senior Geologist Suresh Kumar Rana, Senior Geologist Sanjay Jungral, Senior Geologist	Date of submission of report to the Supervisory Officer:  Signature of the senior most author 29.06.18
Name of Supervisory Officer Sh. Vijay Shivgotra, Director	Date of receipt of report: 29.06.2018 Date of completion of scrutiny: 25/07/2018 Date of handing over report to the author after scrutiny:  Signature
Name of author(s) To whom the report has been handed over by the Supervisory Officer Devender Singh, Senior Geologist Suresh Kumar Rana, Senior Geologist Sanjay Jungral, Senior Geologist	Date of receipt after scrutiny 25/07/2018 Signature.....  Date of resubmission 20/08/2018 Signature..... 
Supervisory Officer Sh. Vijay Shivgotra, Director	Date of receipt after modification 10/08/2018 Date of handing over report to the Regional Mission Head.....10/08/2018 Signature..... 
Name of Regional Mission Head  Deputy Director General	Date of receipt of report 13-08-2018 Date of handing over the report after peer review to the Project Director Signature..... 

Name of Supervisory Officer Sh. Vijay Shivgotra, Director	Date of receipt from the Regional Mission Head after peer review ... <u>12/09/2018</u> Date of handing over to author(s) ... <u>13/09/2018</u> Signature... <u>Vijay Shivgotra</u>
Name of author(s) Devender Singh, Senior Geologist Suresh Kumar Rana, Senior Geologist Sanjay Jungral, Senior Geologist	Date of receipt of report from Supervisory Officer ... <u>13.09.2018</u> Date of handing over report after modification to Supervisory Officer ... <u>20.09.2018</u> Signature... <u>[Signature]</u>
Name of Supervisory Officer Sh. Vijay Shivgotra, Director	Date of receipt of report after modification from the author(s) ... <u>20/09/2018</u> Date of handing over report to Regional Mission Head ... <u>20/09/2018</u> Signature... <u>Vijay Shivgotra</u>
Name of Regional Mission Head <u>Vaisha Ashok Aglaune</u> <u>Deputy Director General</u>	Date of receipt after modification from the Project Director ... <u>25-09-2018</u> Date of handing over the report to the HoD of the Region for approval ... <u>25-09-2018</u> Signature... <u>[Signature]</u>
Name of the HoD of the Region <u>Som Nath Chandel</u>	Date of receipt of report for approval Date of approval... <u>27/09/18</u> Date of sending back report to Project Director Signature of the HoD... <u>[Signature]</u>
Name of the Project Director <u>Vijay Shivgotra</u>	Date of receipt of report after approval of HoD... <u>28/09/2018</u> Date of circulation of report... <u>29/09/2018</u> Signature... <u>Vijay Shivgotra</u>



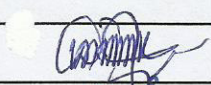
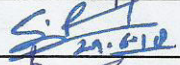
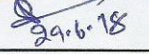
**Geological Survey of India
Certificate of Quality of the report**

**Project/Division: Engineering Geology and Landslide, FSP No: M4-LS/NC/NR/SU-
JK/2017/12224, Mission: IV**

**Title: A note on the Landslide and Slope stability assessment of Shri mata Vaishno Devi
Himkoti track, Reasi district, Jammu and Kashmir.**

Part-A

This is to certify that the following officer(s) have carried out the above mentioned FSP, and are responsible for collection of field data, carrying out the required laboratory analysis, compilation and synthesis of field data, coalition of the data and writing of the report as per the approved guidelines. The duration of field stay of each officer(s) is as against his/her name.

Name(s)/Designation of the Field Officer	Total field stay during the FSP (no. of days)	Signature with date
1. Devender Singh, Senior Geologist	06 days	
2. Suresh Kumar Rana, Senior Geologist	06 Days	
3. Sanjay Jungral, Senior Geologist	06 Days	

Part-B

This is to certify that drilling has been carried out and completed as per the table given below:

Drilling Unit No	Date of commencement of drilling	Date of completion of drilling	Name and Signature of the Director Drilling
NA			

Part-C

This is to certify that geophysical component of the item has been carried out as per the table given below:

Name(s)/Designation of Geophysist	Date of commencement of work	Date of completion of work	Signature with date of the Geophysist
NA			
Name and signature with date of the Director (Geophysics)			

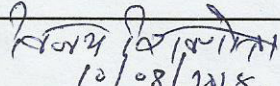
Part-D

This is to certify that the chemical analysis of the geological samples has been carried out as per the table given below:

Name of Director in charge of the Chemical Lab	Dates of receipt of samples from the field officer/party	Date of providing complete analytical results to the field officer	Signature with date of the Director Chemistry
	NA	NA	NA

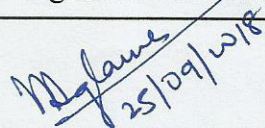
Part-E

This is to certify that the below mentioned Supervisory Officer(s) have supervised the above item and have provided guidance to the Field Officer(s). The number of inspection visits and total duration of field visits is as per the below table. It is also certified that the work carried out under the project is of high standards, the norms of field work, lab work etc. have been followed. It is also certified that thorough scrutiny of the report for maintaining the scientific contents and its Quality has been overseen.

Name(s)/Designation of the Supervisory Officer	No. of inspection visits made to the field area	Total no. of days spent on inspection visit	Signature with date
Sh. Vijay Shivgotra, Director	02	02	 12/08/2018

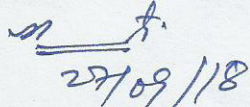
Part-F

This is to certify that the report has been thoroughly scrutinized at the Regional Mission Head level. The scientific contents and the Quality of the report are certified. The report has been peer reviewed and the needful corrections have been attended to.

Name(s)/Designation of the Regional Mission Head	Signature with date
Varsha Ashok Aglawe Deputy Director General	 25/09/2018

Part-G

This is to certify that the above report has been scrutinized at various levels and has also been peer reviewed. On being satisfied regarding the scientific contents and the Quality of the report, the undersigned has accorded approval for its circulation.

Name(s)/Designation of the Regional HOD	Signature with date
Som Nath Chandel, ADG & H.O.D	 27/09/18